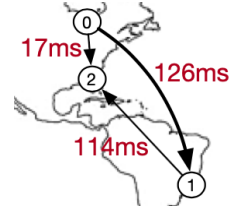


Analysis of FlexCast Reconfiguration

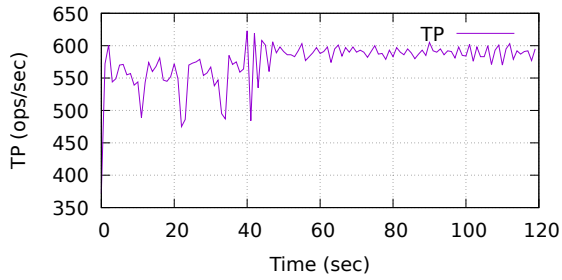
1 Experiment Config.

- TPCC/AWS Latencies
- Regions/Nodes: 3
(Canada, Sao Paulo, Virginia)
- Exp. Duration: 2 min
- Clients: 150 (50 per region)
- Reconfig: 30 seconds
- Locality: 95%

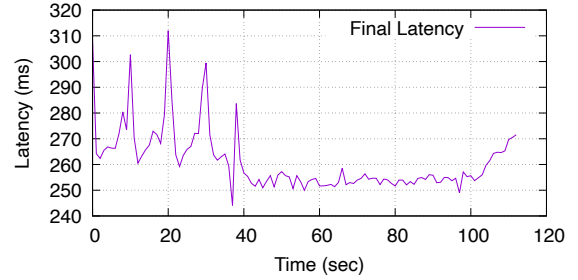


2 Results

2.1 Throughput

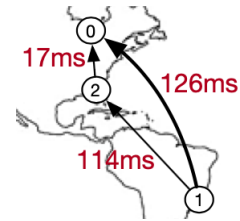


2.2 Latency

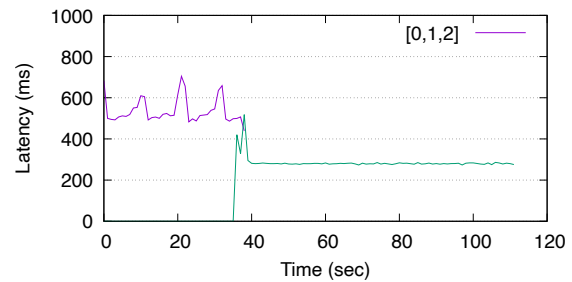
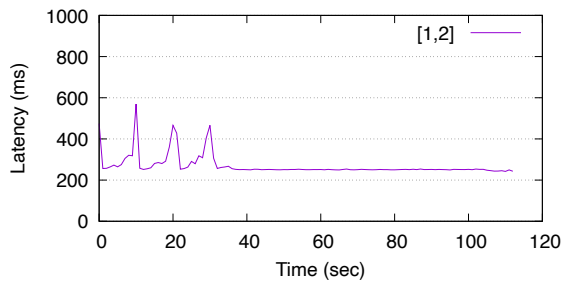
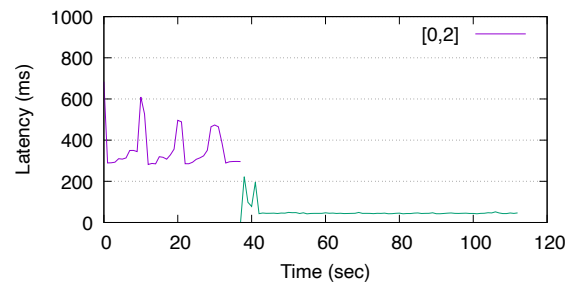
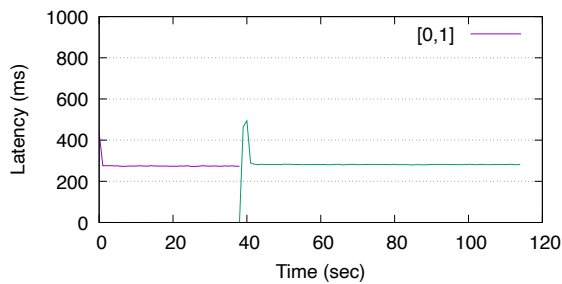


2.3 Reconfiguration details

```
"workload": [  
  {  
    "destination": [0, 1],  
    "percentage": 0.59979254  
  },  
  {  
    "destination": [0, 2],  
    "percentage": 0.032745477  
  },  
  {  
    "destination": [0, 1, 2],  
    "percentage": 0.049208425  
  },  
  {  
    "destination": [1, 2],  
    "percentage": 0.31825358  
  },  
]  
Took 24 ms
```

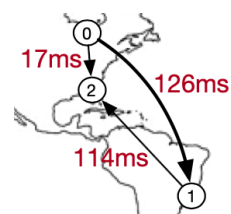


2.4 Latencies per dests (90p)



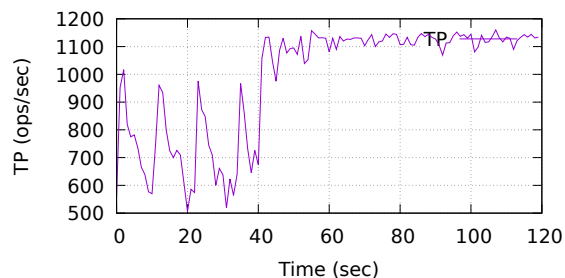
3 Experiment Config.

- TPCC/AWS Latencies
NO CLIENT LATENCIES
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- **GC: 10 seconds**
- **Regions/Nodes: 3**
(Canada, Sao Paulo, Virginia)
- **Clients: 150** (50 per region)
- **Locality: 95%**

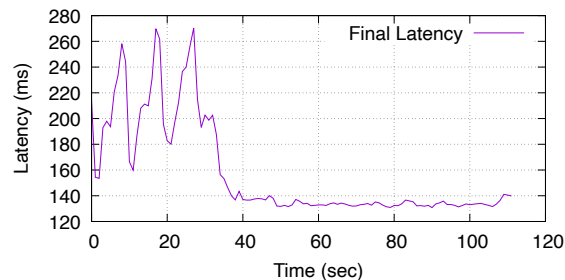


4 Results

4.1 Throughput



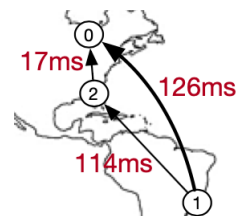
4.2 Latency



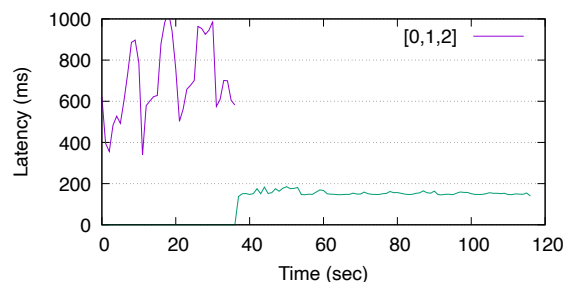
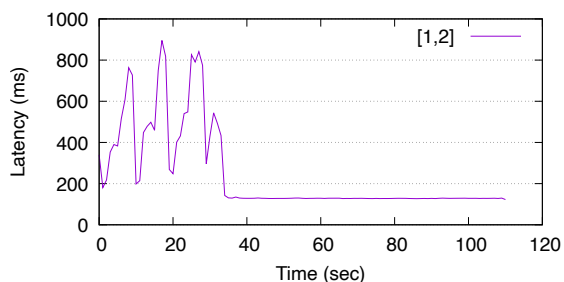
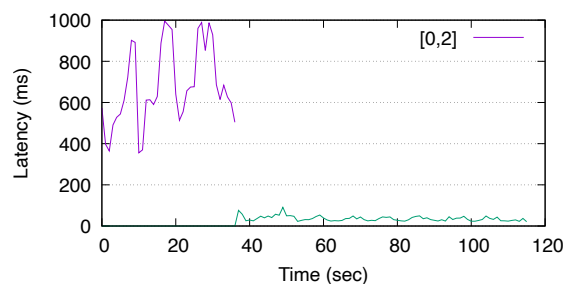
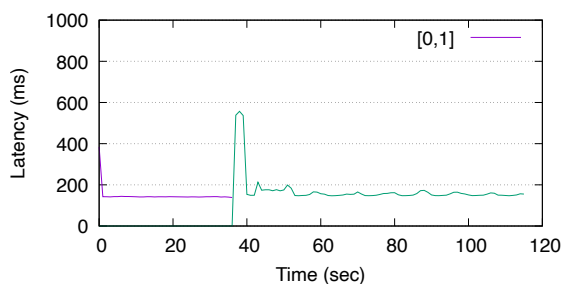
4.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.714834
  },
  {
    "destination": [0, 2],
    "percentage": 0.029263614
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.047801804
  },
  {
    "destination": [1, 2],
    "percentage": 0.2081006
  },
]
```

Took 22 ms

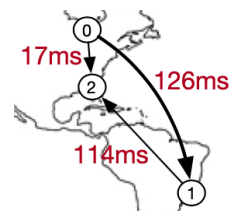


4.4 Latencies per dests (90p)



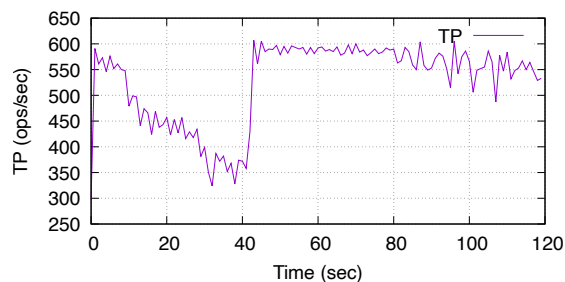
5 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- **NO GC**
- **Regions/Nodes: 3**
(Canada, Sao Paulo, Virginia)
- **Clients: 150** (50 per region)
- **Locality: 95%**

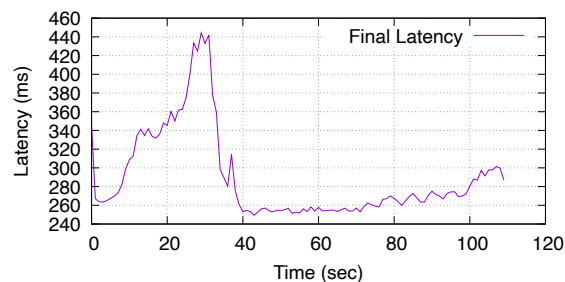


6 Results

6.1 Throughput



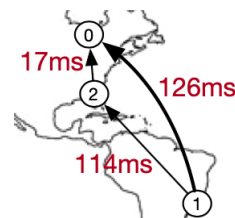
6.2 Latency



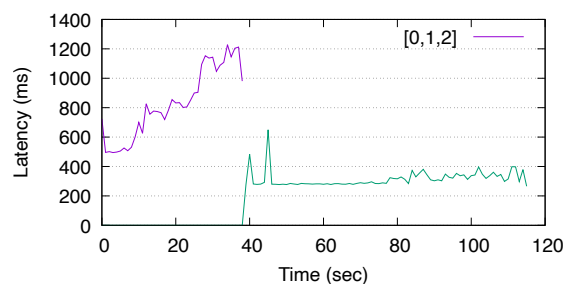
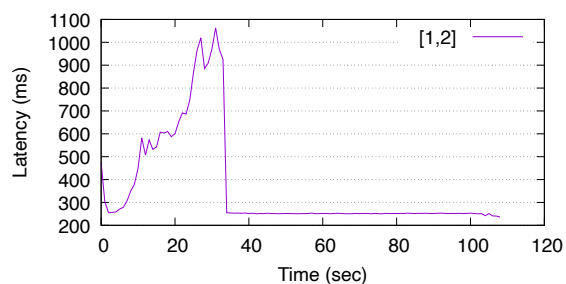
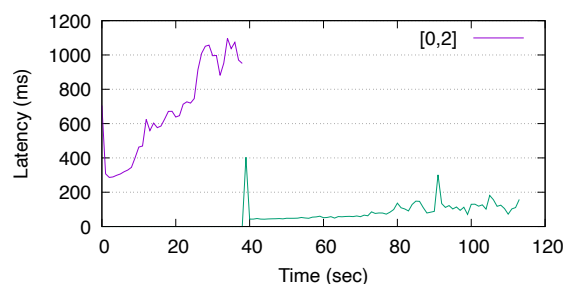
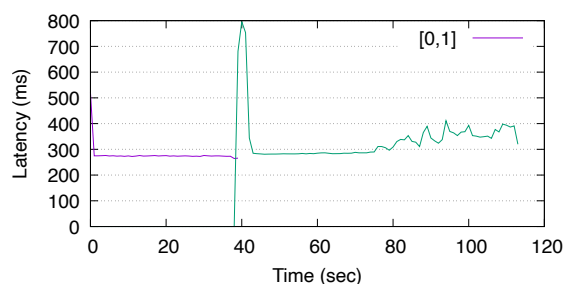
6.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.6664093
  },
  {
    "destination": [0, 2],
    "percentage": 0.029835107
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.05035019
  },
  {
    "destination": [1, 2],
    "percentage": 0.2534054
  }
]
```

Took 25 ms

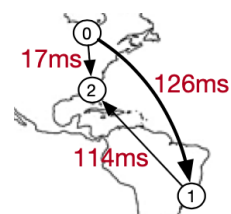


6.4 Latencies per dests (90p)



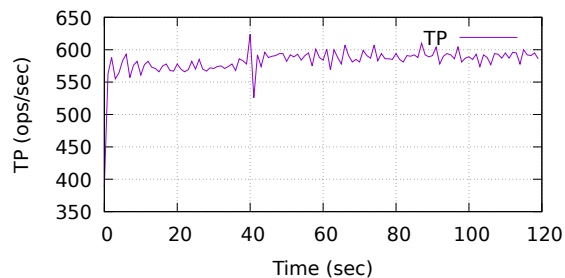
7 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- GC 1 second
- Regions/Nodes: 3
(Canada, Sao Paulo, Virginia)
- Clients: 150 (50 per region)
- Locality: 95%

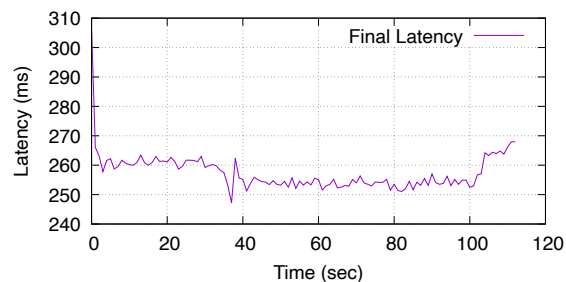


8 Results

8.1 Throughput



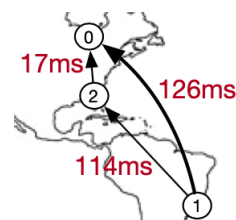
8.2 Latency



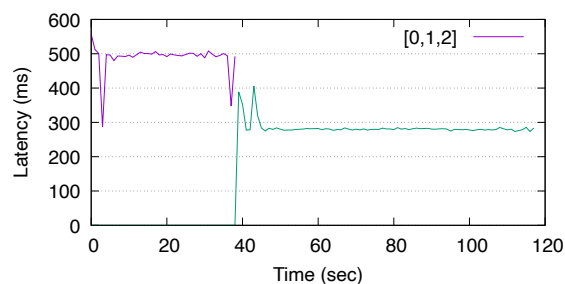
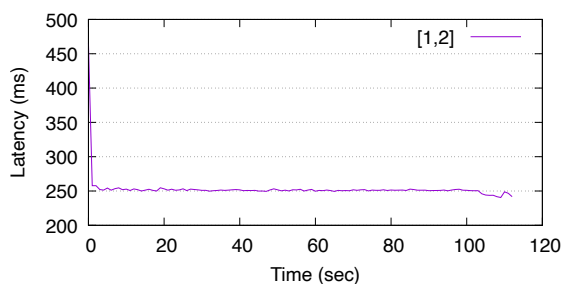
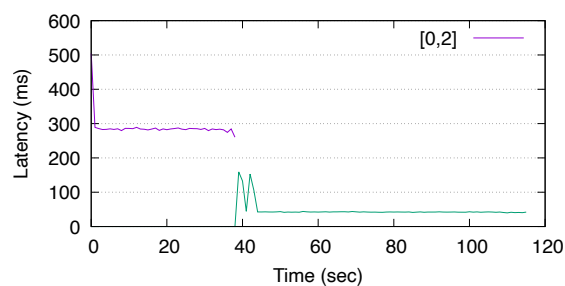
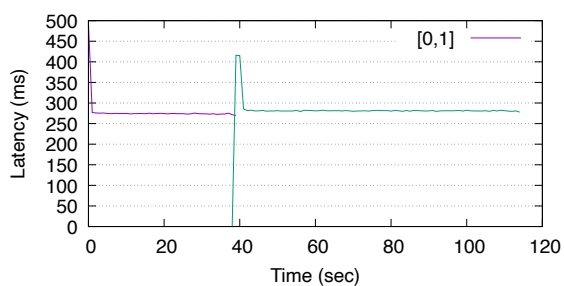
8.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.5822697
  },
  {
    "destination": [0, 2],
    "percentage": 0.031214725
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.050273508
  },
  {
    "destination": [1, 2],
    "percentage": 0.33624208
  },
]
```

Took 22 ms

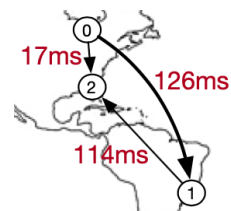


8.4 Latencies per dests (90p)



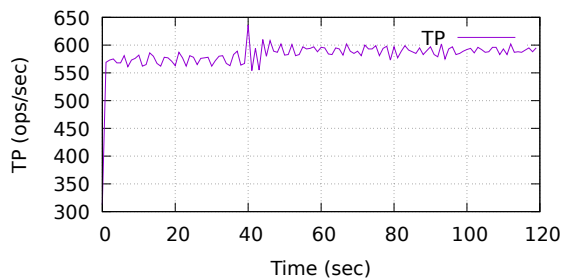
9 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- GC 2 second
- Regions/Nodes: 3
(Canada, Sao Paulo, Virginia)
- Clients: 150 (50 per region)
- Locality: 95%

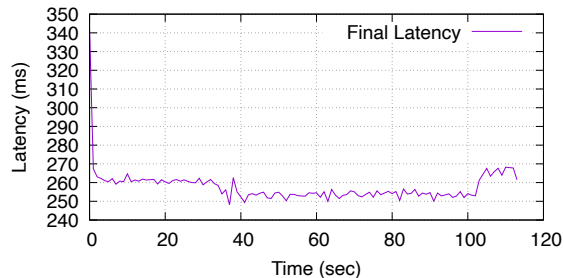


10 Results

10.1 Throughput



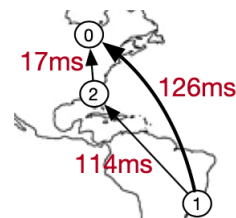
10.2 Latency



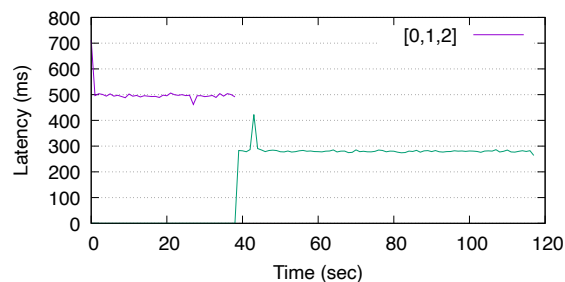
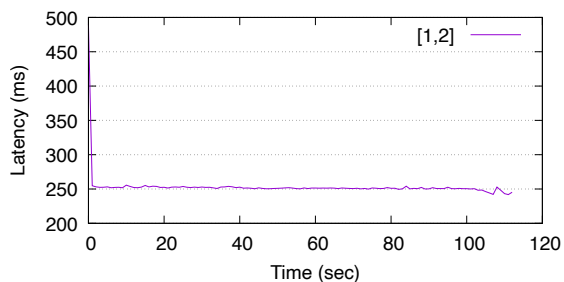
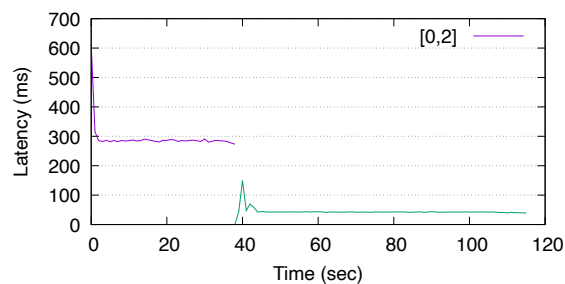
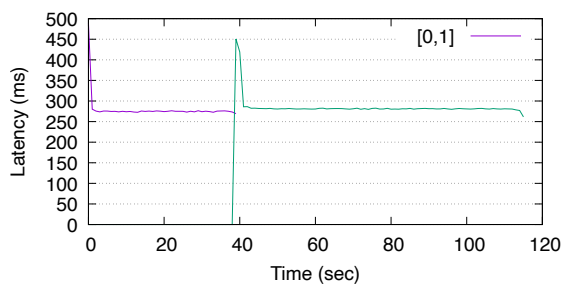
10.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.58594364
  },
  {
    "destination": [0, 2],
    "percentage": 0.03219614
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.04894861
  },
  {
    "destination": [1, 2],
    "percentage": 0.3329116
  }
]
```

Took 22 ms

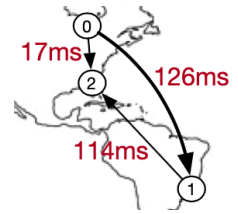


10.4 Latencies per dests (90p)



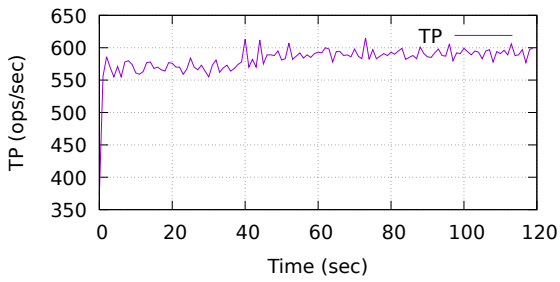
11 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- GC 5 seconds
- Regions/Nodes: 3
(Canada, Sao Paulo, Virginia)
- Clients: 150 (50 per region)
- Locality: 95%

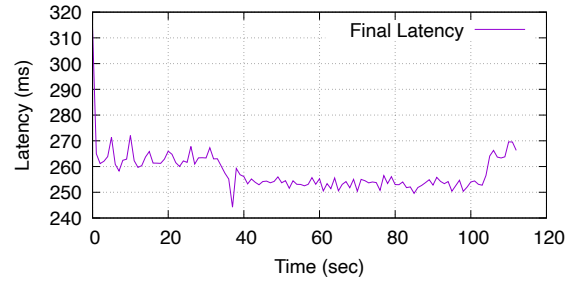


12 Results

12.1 Throughput



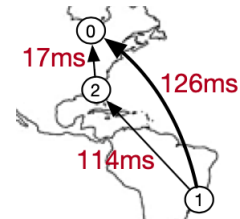
12.2 Latency



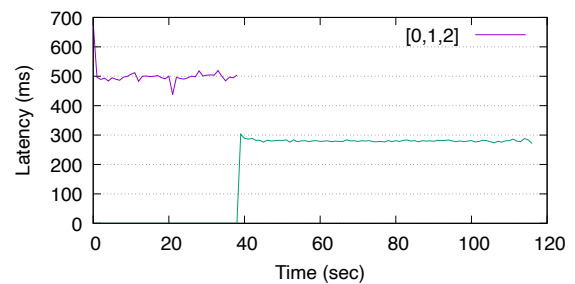
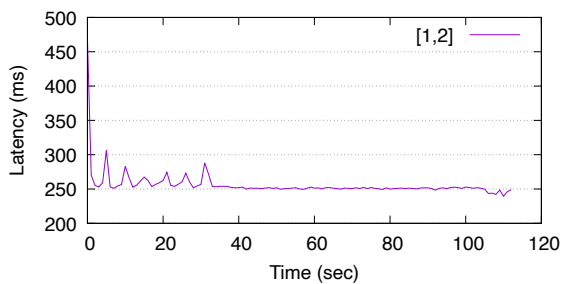
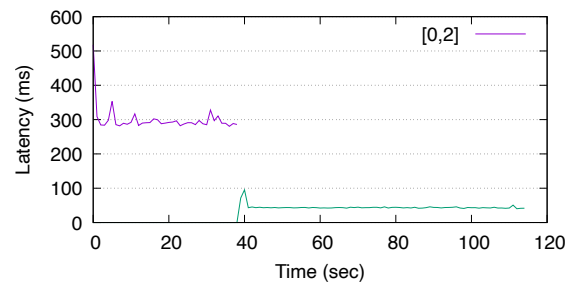
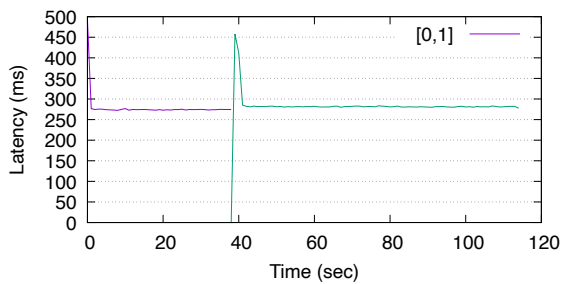
12.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.58656037
  },
  {
    "destination": [0, 2],
    "percentage": 0.03316103
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.049413003
  },
  {
    "destination": [1, 2],
    "percentage": 0.3308656
  },
]
```

Took 22 ms

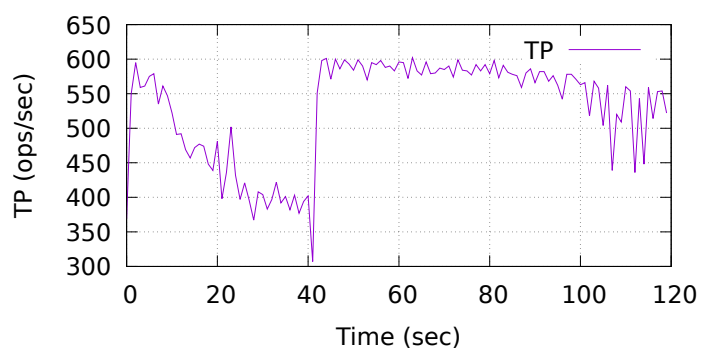


12.4 Latencies per dests (90p)

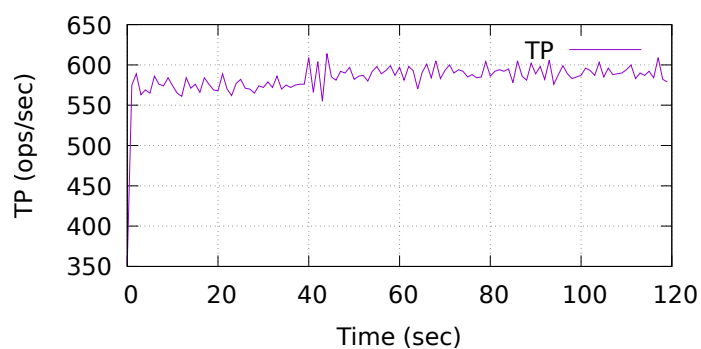


13 Varying gargbage collection freq

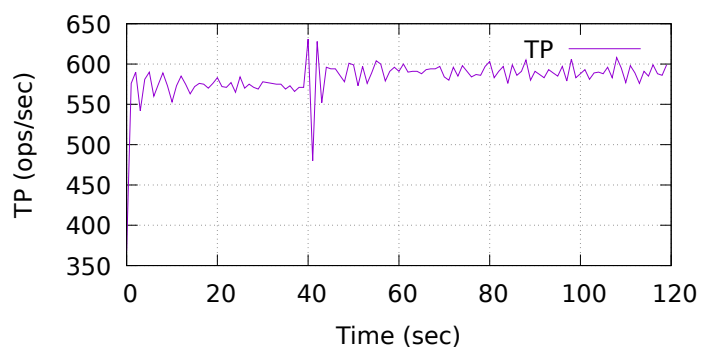
13.1 Throughput



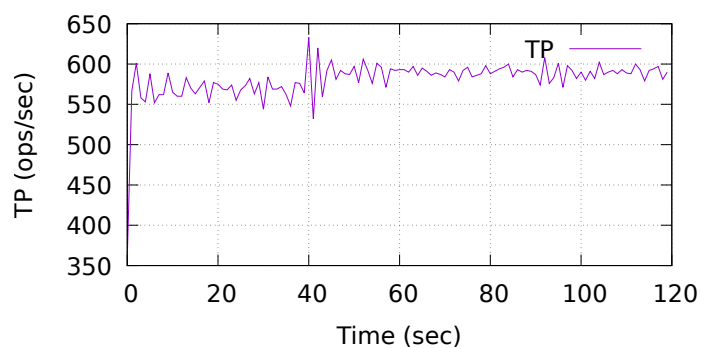
(a) No GC



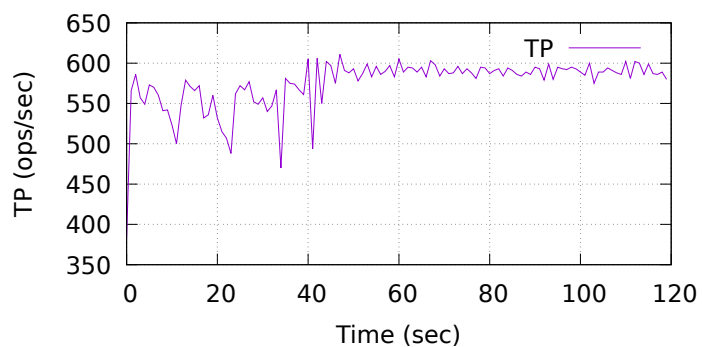
(b) GC 1 second



(c) GC 2 seconds

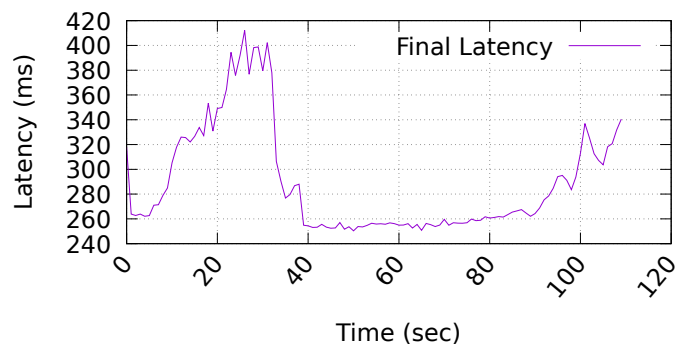


(d) GC 5 seconds

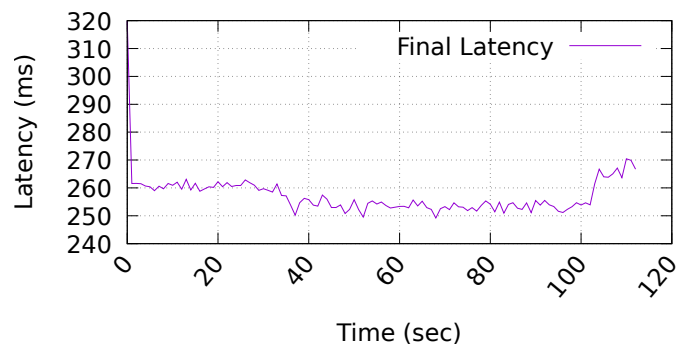


(e) GC 10 seconds

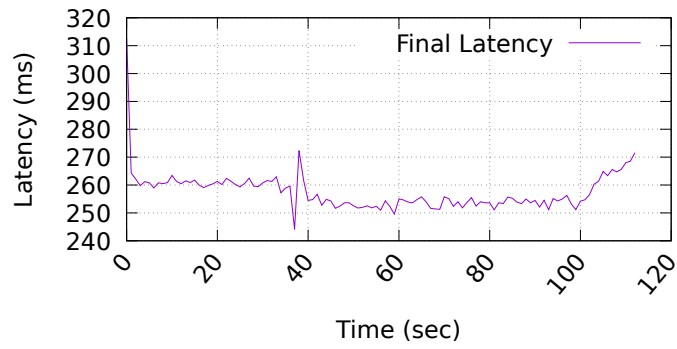
13.2 Latency



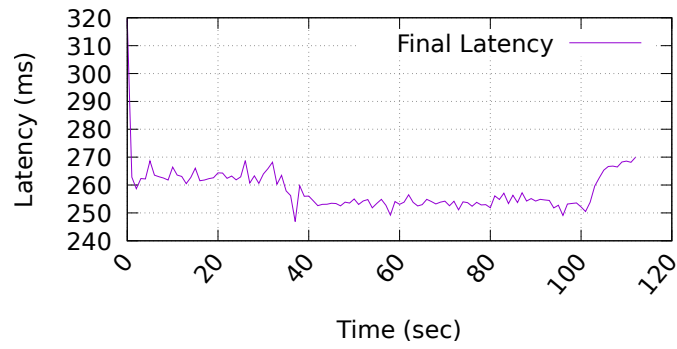
(a) No GC



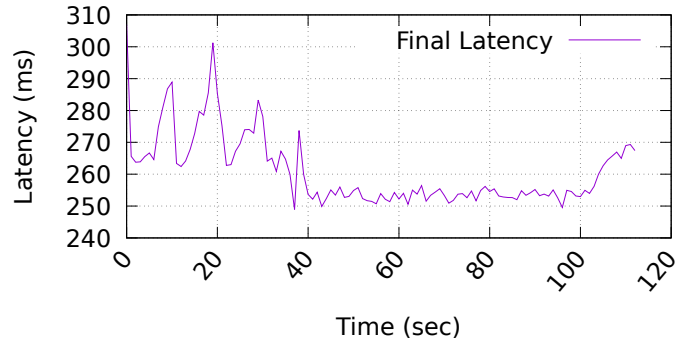
(b) GC 1 second



(c) GC 2 seconds

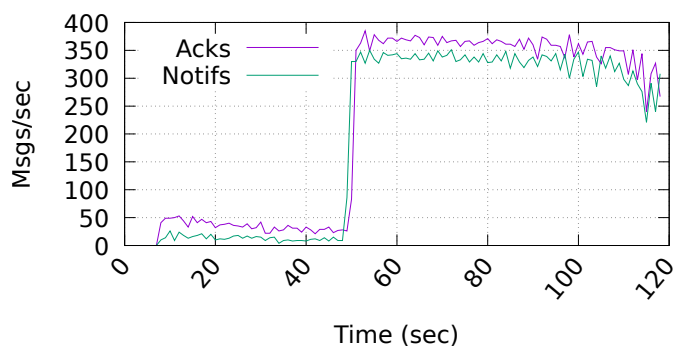


(d) GC 5 seconds

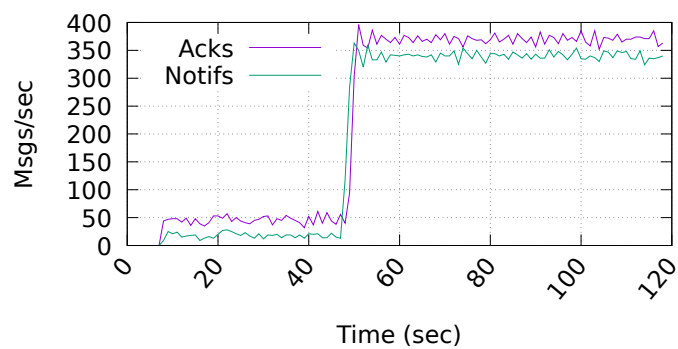


(e) GC 10 seconds

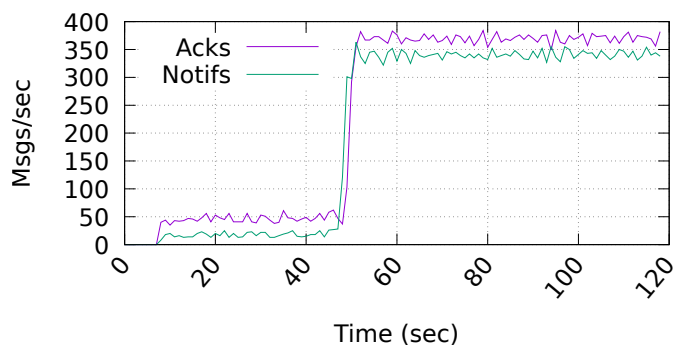
13.3 Volume of acks and notifs



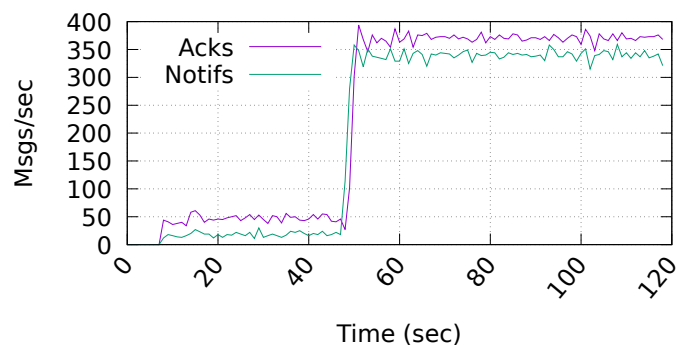
(a) No GC



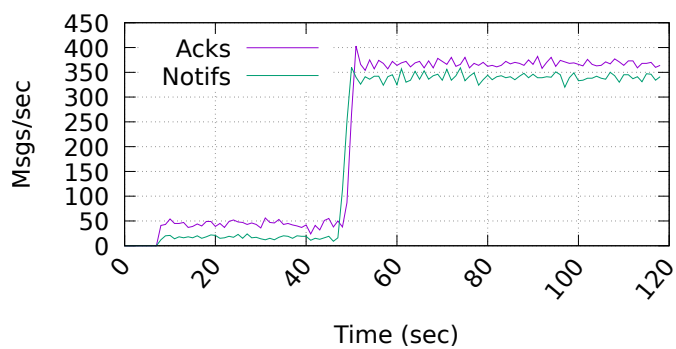
(b) GC 1 second



(c) GC 2 seconds

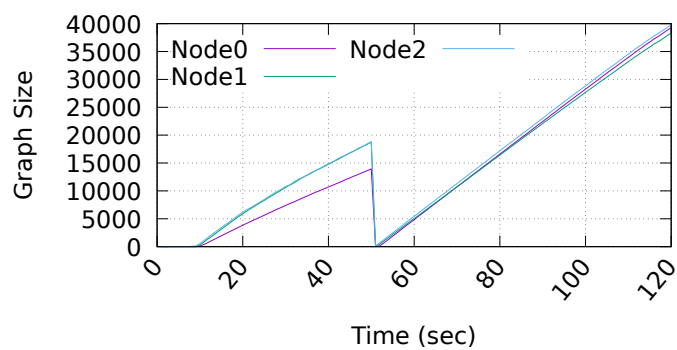


(d) GC 5 seconds

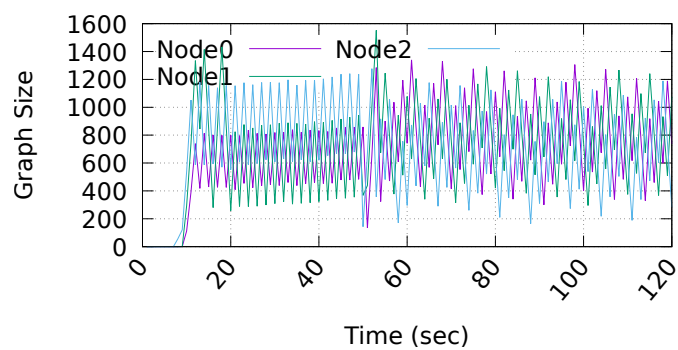


(e) GC 10 seconds

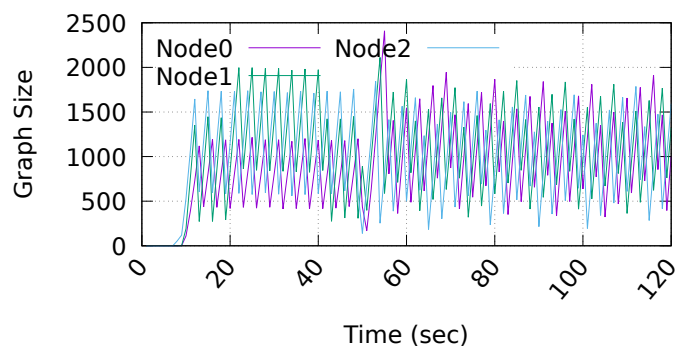
13.4 Size of internal history graph



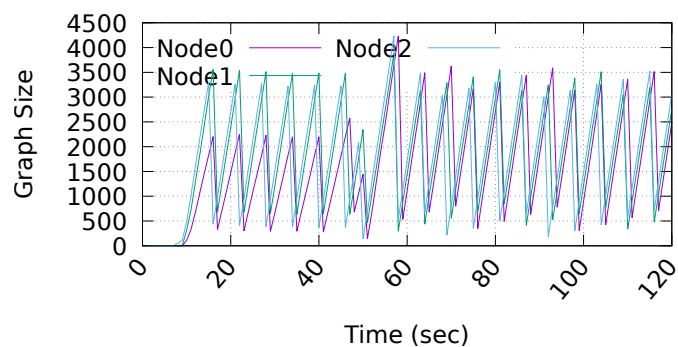
(a) No GC



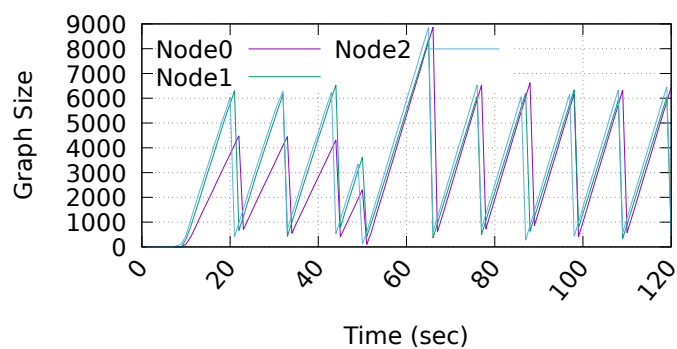
(b) GC 1 second



(c) GC 2 seconds

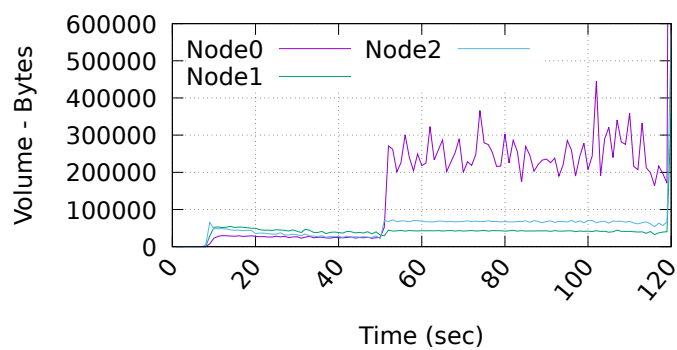


(d) GC 5 seconds

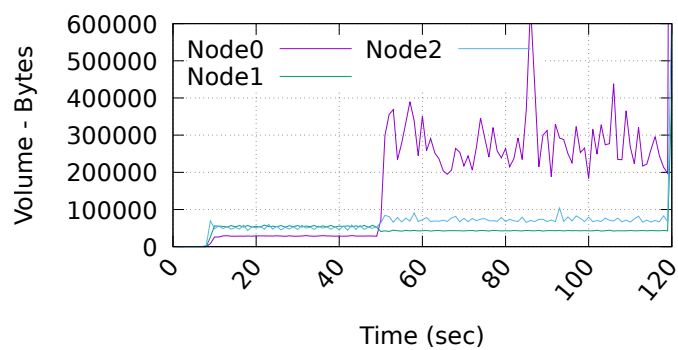


(e) GC 10 seconds

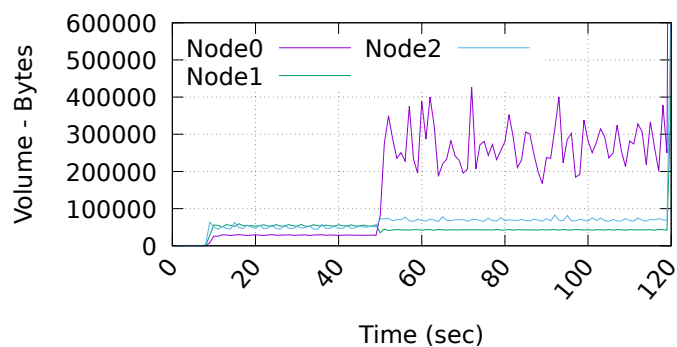
13.5 Volume of data in messages



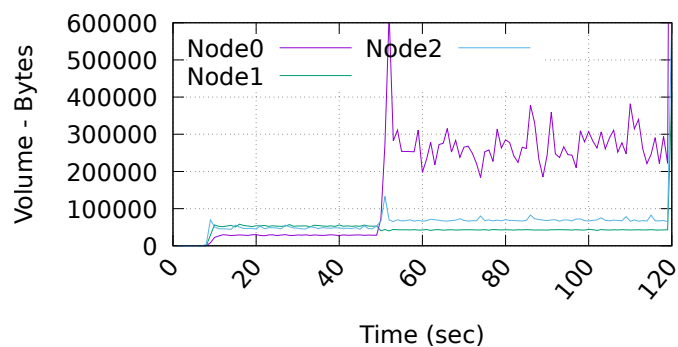
(a) No GC



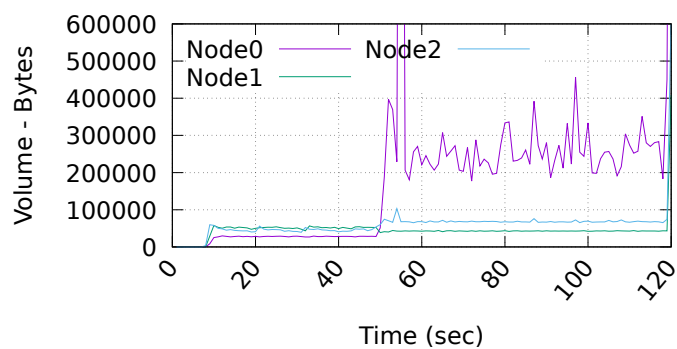
(b) GC 1 second



(c) GC 2 seconds



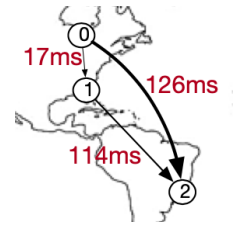
(d) GC 5 seconds



(e) GC 10 seconds

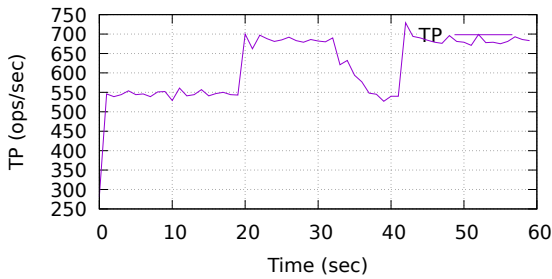
14 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 1 min
- Reconfig: 10 seconds
- GC 1 second
- Regions/Nodes: 3
(Canada, Virginia, Sao Paulo)
- Clients: 150 (50 per region)
- Locality: 95%

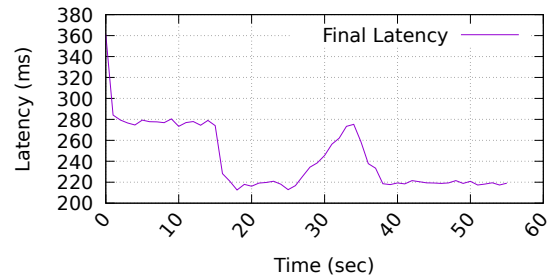


15 Results

15.1 Throughput



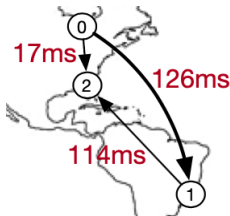
15.2 Latency



15.3 Reconfiguration details

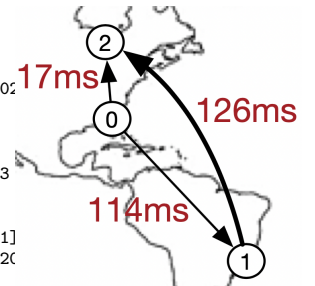
```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.035717566
  },
  {
    "destination": [0, 2],
    "percentage": 0.5288771
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.04792948
  },
  {
    "destination": [1, 2],
    "percentage": 0.38747588
  },
]
```

Took 22 ms



```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.03127702
  },
  {
    "destination": [0, 2],
    "percentage": 0.6091093
  },
  {
    "destination": [0, 2, 1]
    "percentage": 0.04655526
  },
  {
    "destination": [2, 1],
    "percentage": 0.3130585
  },
]
```

Took 22 ms



15.4 Latencies per dests (90p)

